

Decreasing Universe Theory: Black Holes

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Abstract

The Decreasing Universe Theory (DUT) postulates that the spatial metric contracts continuously under the influence of gravitational fields. In this work, we start from the fundamental contraction equation of the DUT, valid instantaneously for any gravitational configuration, and show that its naive application to a stellar core would lead to collapse on a timescale of hours. We investigate a retardation mechanism based on diffusive transport, in which the effective collapse is limited by gravitational diffusion of the plasma, which replenishes the inner layers and keeps the density approximately uniform. The total collapse time of the solar core, governed by this coupled mechanism, is of order $\tau \sim 10^{10}$ years (tens of billions of years), consistent with the observed longevity of the Sun.

1. Introduction

The Decreasing Universe Theory (DUT) proposes that gravitational contraction acts on all massive bodies as a direct consequence of the metric contraction of space. The fundamental equation of the theory describes the rate of change under a gravitational field. However, the naive application of this equation to the stellar interior leads to a collapse time of only a few hours, which is clearly inconsistent with observational astrophysics.

This work investigates the hypothesis that the resolution of this apparent paradox lies in the coupling between two processes:

- **Local metric contraction** — fast, governed by the DUT;
- **Gravitational diffusion of the plasma** — slow, transporting mass between layers to keep the density uniform.

The effective collapse time is determined by the slower process — diffusion — which operates on timescales of billions of years.

Methodological note: all collapse criteria used in this article (Schwarzschild radius $R_S = 2GM/c^2$, Schwarzschild density ρ_S , and the Buchdahl limit) are derived from the standard field equations of General Relativity, not from the DUT's own equations. Should the DUT eventually require an extension of these field equations — just as it already modifies the interpretation of cosmological expansion — these criteria, and all collapse times derived from them, may need complete revision. The quantitative results in this article should therefore be read as estimates conditional on this working hypothesis, not as inevitable consequences of the DUT. This caveat is revisited in more detail in Section 6.2.

1.1. The Buchdahl Limit

It is worth noting that the **Buchdahl theorem**, a rigorous result of General Relativity, establishes that no spherical body in hydrostatic equilibrium with non-negative energy density can have a radius smaller than $(9/8)R_S$ [Schwarzschild radius]. **In the standard model, collapse into a black hole requires the violation of this equilibrium** — typically through the exhaustion of supporting pressure. The DUT offers a distinct conceptual way out: the physical measured radius of the object contracts progressively through metric contraction, and may cross the fixed value without the local hydrostatic equilibrium needing to be violated.

2. The Fundamental Contraction Equation of the DUT

2.1. Derivation of the Contraction Equation

The DUT postulates that any length L under a constant gravitational field g contracts exponentially with time:

$$L(t) = L_0 \cdot \exp(- (J_0/G) \cdot g \cdot t) \quad (F0)$$

where J_0 is the Jocaxian Constant and G is the gravitational constant. Differentiating (F0) with respect to time:

$$dL/dt = - (J_0/G) \cdot g \cdot L \quad (F1)$$

which can be rewritten in dimensionless form:

$$(dL/dt) / L = - (J_0/G) \cdot g \quad (F2)$$

This is the fundamental instantaneous law of the DUT: the relative contraction rate is proportional to the local gravitational field at that instant. It holds even when g varies with time, since it expresses a local differential relation — independent of the system's history, depending only on the instantaneous state.

Making g variable — for a sphere of mass M and radius R , the gravitational field at the surface is $g = GM/R^2$. Substituting into (F2) with $L = R$:

$$(dR/dt) / R = -(J_0/G) \cdot (GM/R^2) = -J_0 \cdot M / R^2$$

Therefore:

$$dR/dt = -J_0 \cdot M / R \quad (1)$$

This is the fundamental contraction equation of the DUT for a massive sphere, derived directly from the exponential law (F0) by differentiation and substitution of the local gravitational field.

Measurement postulate: throughout this work, "contraction" refers to the contraction of matter and of the measuring instruments — atoms, nuclei, rulers — not of space itself, which remains an unchanged vacuum in the DUT. Density increases because the volume occupied by the same amount of matter decreases, not because mass is added to the system. Additionally, J_0 is not an absolute universal constant, but a calibration ratio: it expresses the contraction rate of a given gravitational field relative to the terrestrial reference field, which serves as the measurement standard. J_0 thus allows comparison of contraction rates across different gravitational environments — solar, galactic, etc. — always from the terrestrial measurement frame.

2.2. Physical postulate

From (1) we can postulate that the metric contraction rate of a spherical body of radius $R(t)$ and mass $M(t)$ is given by the ordinary differential equation:

$$dR/dt = -J_0 \cdot M(t) / R(t) \quad (1B)$$

where J_0 is the Jocaxian Constant. This equation is a physical postulate of the theory, analogous to Newton's law of cooling: the rate of change is proportional to the "driving force" (in this case, the gravitational field, implicitly contained in the constant J_0) at that exact instant.

Crucial observation: Eq. (1) is valid instantaneously for any configuration, even when M and R vary with time. It is not an integrated solution for stationary conditions, but a local differential law — valid at every instant, regardless of how the system evolves.

2.3. Solution for constant mass

If the enclosed mass does not depend on the radius ($M = M_0$), Eq. (1B) separates directly:

$$R \cdot dR = -J_0 \cdot M_0 \cdot dt$$

Integrating from R_0 to R :

$$(R^2 - R_0^2) / 2 = -J_0 \cdot M_0 \cdot t$$

$$R(t) = \sqrt{(R_0^2 - 2 \cdot J_0 \cdot M_0 \cdot t)} \quad (2)$$

The total collapse time ($R \rightarrow 0$) is:

$$\tau_{\text{DUT}} = R_0^2 / (2 \cdot J_0 \cdot M_0) \quad (3)$$

2.4. The fast-collapse problem

Substituting the solar-core values ($R_0 = 1.176 \times 10^8$ m, $M_0 = 3.592 \times 10^{29}$ kg, $J_0 = 1.37 \times 10^{-18}$ m² s⁻¹ kg⁻¹):

$$\tau_{\text{DUT}} \approx 3.9 \text{ hours}$$

This result is physically absurd: the Sun's core would disappear in less than a day. The resolution of this paradox does not lie in modifying Eq. (1B), which is correct as an instantaneous law, but in recognizing that the layers of the core do not contract independently — **they are coupled by the plasma, which must redistribute itself to maintain uniform density.**

3. The Uniform-Density Layer Model

3.1. Structure of the stellar core and coupled equations

We divide the solar core into concentric spherical shells. Let $R(t)$ be the radius of the outermost layer (the core surface) and $r(t)$ the radius of any inner layer, with total mass M_0 conserved. The density is kept approximately uniform by plasma transport (gravitational diffusion). The system is described by four coupled equations:

$$dr/dt = -J_0 \cdot M(r) / r \quad (4a)$$

$$M(r) = \rho(R) \cdot (4\pi/3) \cdot r^3 \quad (4b)$$

$$\rho(R) = M_0 / [(4\pi/3) \cdot R^3] \quad (4c)$$

$$dR/dt = -J_0 \cdot M_0 / R \quad (4d)$$

Eq. (4d) describes the contraction of the outer layer and is formally identical to Eq. (1) with constant M . Eqs. (4a–4c) describe how each inner layer contracts as a function of the instantaneous density $\rho(R)$, which in turn depends only on the outer radius $R(t)$.

3.2. Solution for the outer radius

Since Eq. (4d) is identical to Eq. (1) with constant mass M_0 , its solution was already obtained in Section 2.3:

$$R(t) = \sqrt[3]{(R_0^3 - 2 \cdot J_0 \cdot M_0 \cdot t)} \quad (5)$$

with collapse time ($R \rightarrow R_s$):

$$\tau_{\text{DUT}} = (R_0^3 - R_s^3) / (2 \cdot J_0 \cdot M_0) \approx 3.9 \text{ hours} \quad (6)$$

3.3. Solution for inner layers

Substituting (4b) and (4c) into (4a), the equation for an inner layer becomes $dr/dt = -J_0 \cdot M_0 \cdot r^2/R(t)^3$, which can be integrated analytically using the known solution (5) for $R(t)$. The result is the closed-form solution:

$$1/r(t) = 1/r_0 + 1/R(t) - 1/R_0 \quad (7)$$

where r_0 is the initial position of the layer (at $t=0$, when $r_0 = R_0$ one recovers $r(t) = R(t)$, as expected).

3.4. Asymptotic convergence of the layers

Analyzing (7) in the limit $R(t) \rightarrow R_s$, with $R_s \ll R_0 \ll r_0$ for any non-infinitesimally central layer, the term $1/R(t)$ comes to dominate over $1/r_0$ and $1/R_0$. The result is:

$$r(t) \rightarrow R(t) \quad \text{when } R(t) \rightarrow R_s$$

In other words, all layers of the core converge to the same radius simultaneously in the final stages of collapse — the core collapses as an approximately rigid body, even though each layer evolves at its own rate over most of the process. This justifies, a posteriori, treating the collapse as governed by a single effective radius $R(t)$, as done in the following sections.

Numerical result: for a layer at $r_0 = 0.5 \cdot R_0$, when $R(t)$ reaches R_s , the same layer is already at $r(t) \approx R_s$ — the relative difference between $r(t)$ and $R(t)$ is of order $R_s/r_0 \sim 10^{-5}$, numerically negligible.

3.5. Persistence of the fast-collapse problem

The four coupled equations (4a–4d) are mutually consistent and correctly describe the internal structure of the core. However, the collapse time of the outer radius — which governs the collapse of the system as a whole, by the asymptotic convergence of Section 3.4 — remains the same as in Eq. (6): only ~3.9 hours. The retardation mechanism is still needed and is developed in Section 4 through coupling with gravitational diffusion of the plasma.

4. The Retardation Mechanism: Gravitational Diffusion

4.1. Physics of the process

Pure DUT contraction attempts to increase the density rapidly. However, if the outer layers were to contract freely, they would create a prohibitive density gradient. To keep the density approximately uniform, the plasma must redistribute itself: heavy ions sink and light ions rise. This mass transport is governed by the diffusion equation:

$$\partial \rho / \partial t = D \cdot \nabla^2 \rho \quad (8)$$

where $D \approx 9 \times 10^{-4} \text{ m}^2/\text{s}$ is the gravitational diffusion coefficient of helium in the solar plasma (Michaud & Proffitt, 1993).

4.2. Diffusion timescale

The characteristic time for an ion to travel a distance δ by diffusion is:

$$\tau_{\text{diff}} \sim \delta^2 / D \quad (9)$$

For the entire radius of the core ($\delta = R_0 \approx 1.2 \times 10^8 \text{ m}$):

$$\tau_{\text{diff}}^{(R_0)} \sim (1.2 \times 10^8)^2 / (9 \times 10^{-4}) \approx 1.5 \times 10^{19} \text{ s} \approx 487 \text{ billion years}$$

For a shell of thickness $\delta = R_0/4 \approx 2.9 \times 10^7 \text{ m}$:

$$\tau_{\text{diff}}^{(\delta)} \sim (2.9 \times 10^7)^2 / (9 \times 10^{-4}) \approx 9.6 \times 10^{17} \text{ s} \approx 30.5 \text{ billion years}$$

4.3. The coupled model

The effective collapse is determined by the slower process. Pure DUT contraction would take days/years, but diffusion takes billions of years:

$$\tau_{\text{total}} \approx \max(\tau_{\text{DUT}}, \tau_{\text{diff}}) = \tau_{\text{diff}} \quad (10)$$

The approximation $\tau_{\text{total}} \approx \max(\tau_{\text{DUT}}, \tau_{\text{diff}})$ corresponds to the adiabatic elimination technique (quasi-steady-state approximation), widely used in dynamical systems with strongly time-scale-separated processes. Since $\tau_{\text{diff}}/\tau_{\text{DUT}} \sim 10^{15}$ (Section 6.1), the error introduced by this approximation is of the same order, $\sim 10^{-15}$ — negligible compared to any other uncertainty in the model. Fully solving the coupled system of ODEs would not perceptibly change the final result.

The density evolves at the rate limited by diffusion:

$$d\rho/dt \approx \rho / \tau_{\text{diff}} \quad (11)$$

Integrating up to the Schwarzschild density ρ_s :

$$\tau_{\text{collapse}} = \tau_{\text{diff}} \cdot \ln(\rho_s/\rho_0) \quad (12)$$

where $\rho_0 = 5.27 \times 10^4 \text{ kg/m}^3$ and $\rho_s = 5.65 \times 10^{20} \text{ kg/m}^3$, so that $\ln(\rho_s/\rho_0) \approx 36.9$.

The model above represents a first-order estimate. Two additional effects, ignored in the static model, **accelerate** the process and reduce the collapse time:

- **Effect 1 — $\delta(t)$ decreases with $R(t)$:** as the core contracts, $\delta = R(t)/4$ decreases, reducing $\tau_{\text{diff}}(t) = \delta^2/D \propto R(t)^2$. Diffusion becomes progressively faster, accelerating the process in the final stages.
- **Effect 2 — Contraction of ions during transit:** ions in transit (rising or sinking) are not at the edge of the core — they are in interior regions where $g(r) < g_{\text{edge}}$. By the DUT principle, they contract during their own journey, accumulating additional contraction. For uniform density, $g(r) = GM \cdot r/R^3$, and the average along the path from 0 to R is $\langle g \rangle = g_{\text{edge}}/2$. This corresponds to an acceleration factor of $\sim 1.5\times$ on the rate of density increase.

Numerical integration of the coupled model, incorporating both effects, gives:

$$\tau_{\text{collapse}} \text{ (coupled)} \approx 30 \text{ billion Years}$$

4.4. Limitations of the Diffusion Mechanism

The gravitational diffusion mechanism described in this section is a first-order idealization. In a self-gravitating fluid in hydrostatic equilibrium, a density fluctuation tends to be corrected not by diffusive transport, but by pressure waves, on a timescale given by the sound-crossing time:

$$\tau_{\text{sound}} \sim R_0/c_s \sim 10^8 \text{ m} / 10^5 \text{ m/s} \sim 10^3 \text{ s} \sim 15\text{--}30 \text{ minutes}$$

This scale is many orders of magnitude faster than molecular diffusion ($\sim 10^{19} \text{ s}$). This means that diffusion, as used here, correctly describes the transport of chemical composition (helium/hydrogen ratio) — but not necessarily the mechanism that prevents the total density from re-equilibrating under the metric contraction of the DUT. The effective retardation of the collapse may instead depend on a modification of the hydrostatic dynamics or of the effective equation of state under continuous metric contraction — whose complete formulation from the principles of the DUT remains an open problem for future work.

5. Numerical Results

5.1. Solar core parameters

Parameter	Value
Core radius (R_0)	$1.176 \times 10^8 \text{ m} = 0.169 R_\odot$
Core mass (M_0)	$3.592 \times 10^{29} \text{ kg} = 0.181 M_\odot$
Schwarzschild radius (R_s)	533.5 m
Initial density (ρ_0)	$5.27 \times 10^4 \text{ kg/m}^3$
Schwarzschild density (ρ_s)	$5.65 \times 10^{20} \text{ kg/m}^3$
Diffusion coefficient (D)	$9 \times 10^{-4} \text{ m}^2/\text{s}$
Jocaxian Constant (J_0)	$1.37 \times 10^{-18} \text{ m}^2 \text{ s}^{-1} \text{ kg}^{-1}$

5.2. Compared timescales

Process	Collapse time
Pure DUT contraction ($M = \text{constant}$)	$\sim 4 \text{ hours}$
Coupled equations (4a–4d), outer layer	$\sim 3.9 \text{ hours (identical)}$
Static model ($\delta = R_0/4$ fixed, no coupling)	$\sim 1.125 \times 10^{12} \text{ years (over a trillion)}$
+ Effect 1: $\delta(t) = R(t)/4$ variable	$\sim 46 \text{ billion years}$
+ Effect 2: ions contract during transit	$\sim 30 \text{ billion years} \checkmark$
Age of the Sun (observed)	$\sim 4.6 \text{ billion years}$
Age of the universe (DUT via CMB: $t = \ln(1+z)/H_0$)	$\sim 97 \text{ billion years}$

5.3. Effective collapse time

With the choice of shell thickness $\delta = R_0/4$:

$$\tau_{\text{collapse}} \approx 30 \text{ billion years}$$

This result is of order tens of billions of years, consistent with the age of the universe according to the DUT and ensuring that the Sun still has billions of years of stable life ahead.

6. Discussion

6.1. Why is diffusion the bottleneck?

The ratio between the timescales is:

$$\tau_{\text{diff}} / \tau_{\text{DUT}} \sim (R_0^2 / D) / (R_0^2 / (J_0 \cdot M)) = J_0 \cdot M / D \approx 10^{15}$$

This means that DUT contraction acts continuously on the ions emerging from the deeper layers, progressively contracting them. When a contracted ion reaches a density greater than that of the layer just below it, it sinks (because g in the layer above is larger and contracts it faster) — and a less dense ion emerges to take its place. This cycle repeats continuously, and it is this reorganization process — not the

contraction alone — that accounts for most of the total time and that keeps the plasma at an approximately uniform density.

6.2. The Buchdahl limit and the DUT

The Buchdahl theorem, a rigorous result of General Relativity, establishes that no spherical body in hydrostatic equilibrium with non-negative energy density can have a radius smaller than $(9/8)R_s$. In the standard model, collapse into a black hole requires the violation of this equilibrium — typically through the exhaustion of supporting pressure.

The DUT offers a distinct conceptual way out: the physical measured radius of the object contracts progressively through metric contraction, and may cross the fixed value R_s without the local hydrostatic equilibrium needing to be violated. This calls into question the standard mechanism of black-hole formation through direct gravitational collapse, and suggests that *the DUT naturally resolves the Buchdahl impasse*.

Caveat: the quantitative results derived here (including the Schwarzschild radius R_s itself) assume the standard General Relativity metric. Just as the DUT reformulates the interpretation of cosmological expansion without discarding General Relativity, it may eventually require an extension of Einstein's field equations. Should this occur, the quantitative results presented in this article — including R_s and the Buchdahl limit itself — may need revision. This possibility is acknowledged as a limitation of the present work.

6.3. Implications for the black-hole population

If the nuclear collapse time is of order tens of billions of years, low- and medium-mass stars have not yet had time to collapse in the current universe ($\tau_{\text{universe}} \approx 97 \times 10^9$ years, according to the DUT estimate from the CMB redshift: $t = \ln(1+z_{\text{CMB}})/H_0$). This restricts the formation of stellar black holes primarily to:

- Massive stars ($M \gtrsim 20 M_\odot$), where R_s is larger and the diffusion time proportionally smaller;
- Objects in environments with extreme gravitational fields (galactic nuclei, globular clusters).

6.4. Sensitivity to the Diffusion Coefficient

The collapse time of ~30 billion years was obtained using the diffusion coefficient $D \approx 9 \times 10^{-4} \text{ m}^2/\text{s}$, corresponding to the microscopic diffusion of helium in solar plasma (Michaud & Proffitt, 1993). As discussed in Section 4.4, this value is specific to chemical (compositional) separation under standard solar conditions, and carries significant uncertainty when applied to the problem of total mass redistribution under continuous metric contraction.

Because the collapse time scales exactly as $\tau_{\text{collapse}} \propto 1/D$ — a direct consequence of D entering the governing equation as a pure multiplicative rate constant — the result is highly sensitive to the precise value

adopted. For illustration, if one adopts a value only three times larger, $D \approx 2.7 \times 10^{-3} \text{ m}^2/\text{s}$, still within the order of magnitude of microscopic transport processes in the solar interior, the coupled model yields:

$$\tau_{\text{collapse}} \approx 10 \text{ billion years}$$

This is of the same order as the current age of the Sun. Conversely, a value $D \sim 10^{-4} \text{ m}^2/\text{s}$ — roughly an order of magnitude smaller — would push the collapse time to $\sim 3 \times 10^{11}$ years, well beyond the age of the universe according to the DUT.

This sensitivity demonstrates that the absolute value of the collapse time is not yet firmly constrained by the present model. What the model does establish is the qualitative mechanism: the effective collapse is governed by a slow transport process competing with rapid metric contraction, and the resulting timescale falls in the range of billions of years rather than hours or trillions of years. A more precise determination of the effective transport coefficient under DUT metric contraction — whether from improved helioseismic constraints, laboratory plasma diffusion data, or a first-principles derivation from the DUT field equations (Section 6.2) — will be essential to narrow this uncertainty.

7. Conclusion

We started from the fundamental contraction equation of the DUT, valid instantaneously for any gravitational configuration. Applying it to the solar core with a uniform-density layer model, we showed that:

- Pure metric contraction is extremely fast (timescale of hours to years);
- The effective collapse is limited by gravitational diffusion of the plasma, which operates on timescales of tens of billions of years;
- The collapse time of the solar core, estimated at ~ 30 billion years, is consistent with the observed longevity of the Sun and the age of the universe according to the DUT.

The DUT remains a testable alternative cosmological theory, with quantitative predictions distinct from the standard Λ CDM model.

8. References

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